THE MULTIPLICITY OF A TWO-PERIODIC COMPLEX BUILT FROM A NILPOTENT ENDOMORPHISM VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The counted statement and proof contain no cross reference at all, so the assembled document defines every label it uses and no external reference stub is needed.

Allowed standard prerequisites (editorial).

- Two-periodic and (2,1)-periodic complexes and their multiplicity e_R. Same chapter, Definition environment labelled definition-periodic-complex (source line 110) together with the displayed definition of e_R at source lines 200-205 (label equation-multiplicity-coker-ker, line 203). Supplies the objects M, N, phi, psi, the notation e_R(M, phi^i, phi^{n-i}) and the convention phi^0 = id_M used in the first line of the counted proof; the vanishing argument is the counted content.
- Length is additive in short exact sequences. Algebra, Lemma labelled algebra-lemma-length-additive (canonical Stacks tag 00IV). Not needed by this card, which works with lengths of the explicit modules K_i/t^j M directly; recorded because the sibling lemma of the same section uses it.

Lemma 0.1. *Let R be a ring. Let M be an R-module. Let $\varphi : M \to M$ be an endomorphism and $n > 0$ such that $\varphi^n = 0$ and such that $\mathrm{Ker}(\varphi)/\mathrm{Im}(\varphi^{n-1})$ has finite length as an R-module. Then*

$$e_R(M, \varphi^i, \varphi^{n-i}) = 0$$

for $i = 0, \ldots, n$.

Proof. The cases $i = 0, n$ are trivial as $\varphi^0 = \mathrm{id}_M$ by convention. Let us think of M as an $R[t]$-module where multiplication by t is given by φ. Let us write $K_i = \mathrm{Ker}(t^i : M \to M)$ and

$$a_i = \mathrm{length}_R(K_i/t^{n-i}M), \quad b_i = \mathrm{length}_R(K_i/tK_{i+1}), \quad c_i = \mathrm{length}_R(K_1/t^iK_{i+1})$$

Boundary values are $a_0 = a_n = b_0 = c_0 = 0$. The c_i are integers for $i < n$ as K_1/t^iK_{i+1} is a quotient of $K_1/t^{n-1}M$ which is assumed to have finite length. We will use frequently that $K_i \cap t^jM = t^jK_{i+j}$. For $0 < i < n-1$ we have an exact sequence

$$0 \to K_1/t^{n-i-1}K_{n-i} \to K_{i+1}/t^{n-i-1}M \xrightarrow{t} K_i/t^{n-i}M \to K_i/tK_{i+1} \to 0$$

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By induction on i we conclude that a_i and b_i are integers for $i < n$ and that

$$c_{n-i-1} - a_{i+1} + a_i - b_i = 0$$

For $0 < i < n-1$ there is a short exact sequence

$$0 \to K_i/tK_{i+1} \to K_{i+1}/tK_{i+2} \xrightarrow{t^i} K_1/t^{i+1}K_{i+2} \to K_1/t^iK_{i+1} \to 0$$

which gives

$$b_i - b_{i+1} + c_{i+1} - c_i = 0$$

Since $b_0 = c_0$ we conclude that $b_i = c_i$ for $i < n$. Then we see that

$$a_2 = a_1 + b_{n-2} - b_1, \quad a_3 = a_2 + b_{n-3} - b_2, \quad \ldots$$

It is straightforward to see that this implies $a_i = a_{n-i}$ as desired. $\square$